

Analysis of the Tic-Tac-Toe Reinforcement Learning Homework Assignment

1. Clarity:

- The problem definition is relatively clear in terms of the goal: implementing a reinforcement learning algorithm using the "Value Function" approach to learn the value of each state in Tic-Tac-Toe.

2. Feasibility:

- It is high for a motivated student who has a basic understanding of RL concepts, including Temporal Difference (TD) learning, Value Iteration, and greedy policies.
- If the student is not familiar with the necessary RL concepts or how to modify the existing self-play code, the task may require additional effort to understand and implement the algorithm.

3. Moderate difficulty:

- The assignment involves implementing RL methods like TD learning and requires some understanding of state-value functions and iterative improvement through self-play.

4. Additional Aspects to Consider:

- A key challenge could be in understanding how to fine-tune the learning rate and how it influences the agent's performance over time.
- The second task (playing against an expert) adds a layer of complexity, as you need to integrate a policy from another agent (the teacher) into the learning process.